

Trading System Audit and Build Roadmap

KEI worked example and Specification V3 evidence review

Prepared 10 September 2026

Audience: a trader with basic market knowledge who wants to understand and help build the software.

Main conclusion

The repository contains a substantial research and reporting system. It downloads and stores NSE daily data, builds market and sector context, detects three price patterns, constructs entries and risk levels, produces V3 reports, and runs historical measurements. The test suite passes. It does not yet provide evidence for funded prospective trading.

The KEI price path is real in the cached Yahoo candles: the market traded through the stated 3R and 4R levels after 1 September 2026. The claim that the system found the short prospectively at 12:45 is not supported by the current evidence. The daily base breakdown uses the completed 1 September daily candle, which was unavailable at 12:45. The reported carry score of 62.2 also includes the 60 minute candle that was still forming. With completed information only, the daily detector shows no short setup and carry scores 45.2, below its floor of 60.

Two additional implementation defects prevent the standard replay from validating this short. The main V3 backtest is long only, and its stop order fill logic uses long side comparisons for every direction. The KEI numbers therefore describe a retrospective worked geometry, not a reproducible live alert or broker executed trade.

The practical recommendation is to keep the system in research and paper mode. The first milestone is to make time availability, direction, fills, and immutable event logging correct before running another selection study.

Evidence labels

Label	Meaning
Verified	Directly reproduced from code, local data, generated files, or tests.
Implemented	The code path exists, but this audit did not prove its external service behavior end to end.
Claimed	A repository document states it, but the underlying run record is incomplete or conflicting.
Manual	A person must supply a choice, credential, fill, or operational action.
Planned	Recommended work that is not present in the current implementation.
Not verified	The available files do not contain enough evidence to make the claim.

Document map

1. Chapters 1 through 11 explain the current system in plain language.
2. Chapter 12 audits KEI candle by candle and separates price outcome from signal validity.
3. Part Two lists code, configuration, data, report, and test evidence.
4. Part Three gives a practical build roadmap and fixed evaluation rules for the next experiment.

Part One Current System

1 Complete system overview

The current system is a Python application named Asymmetry Engine. It is decision support. It has no broker order placement endpoint. A daily run normally happens after the NSE cash session, because the official bhavcopy becomes available after the close. A separate V3 monitor exists for market hours, but it refuses funded alerts unless live Upstox data, a current probability model, sufficient forward evidence, and account equity are available.

Stage	Receives	Does	Passes onward
1 Data access	NSE archives, Yahoo charts, optional Upstox, BSE filings and news	Downloads responses with pacing, retries, and cache	Raw files or parsed tables with a source tier
2 Daily storage	NSE cash bhavcopy and delivery file	Parses EQ rows and replaces one date in SQLite	Daily OHLCV, turnover, trades, delivery percent
3 Timeframes	Daily, 60 minute, 15 minute candles	Builds weekly and 120 minute candles	Frames used by structure, carry, and entry logic
4 Calculations	Price, volume, index, sector and catalyst histories	Computes EMA, ATR, ADR, relative strength, regime and structure	Scores and explanatory notes
5 Setup detection	Daily frame	Checks reclaim, high tight flag, and volume backed base breakout in both directions	One best setup per stock
6 Carry test	60 minute frame	Builds 120 minute frame and scores setup, EMA alignment, headroom, volume and location	Pass or fail plus 0 to 100 score
7 Trade geometry	15 minute frame plus daily and weekly context	Builds trigger, entry, structural stop, valid fill band, target and quantity	V3 plan or a hard rejection
8 Selection	Nine module values and market regime	Computes weighted quality score and applies regime floor	Trade, watch, or rejection
9 Replay	Past 15 minute bars and frozen plan	Simulates next bar entry and walks forward to stop, target, or time limit	Outcome, R, excursions and timestamps
10 Reporting	V3Scan or BacktestResult records	Uses deterministic Python templates	Markdown, terminal tables, HTML, JSON or JSONL

Main implementation technologies are Python 3.11 or later, pandas and NumPy for calculations, SQLAlchemy and SQLite for stored daily data, httpx for network access, Typer and Rich for the command line, pydantic settings for configuration, and generated Markdown and HTML for reports. GitHub Actions and Windows Task Scheduler are the two available scheduling paths.

2 Where the data comes from

Input	Provider and access	Current use	Limits found in the repository
NSE daily stocks	Official nsearchives.nseindia.com zipped cash bhavcopy	Daily OHLCV, turnover and trade count; SQLite backbone	End of day only; no explicit corporate action adjustment
Delivery data	Official NSE MTO file	Delivery percent and delivery plus turnover spike	May be missing; stored as null when unavailable
NIFTY options	NSE F and O bhavcopy or live Upstox option chain	Dealer gamma context	EOD archive or optional live; gamma is context, not entry
Intraday stock bars	Yahoo chart API fallback or authenticated Upstox	15 minute entry and 60 minute carry	Yahoo is delayed and unofficial; current cache holds about 60 calendar days of 15 minute history
NIFTY VIX and global series	Yahoo chart API	Regime and macro context	Availability varies; missing series are omitted
Sector classes	NIFTY constituent Industry labels	Maps stock to sector	Historical snapshot is liquidity based and contains no historical membership
Sector performance	Equal weighted composites built from stored stocks	Sector leadership and stock versus sector RS	Requires at least four constituents
Catalysts	BSE filings, attached results PDFs, and current RSS news	Optional LLM or fixed rule scoring	RSS cannot be backfilled; local catalyst history is incomplete

An API request is an HTTP call to a data service. The code sends a symbol, requested range, and interval. A shortened Yahoo shaped response looks like this:

```
GET /v8/finance/chart/KET NS?range=60d&interval=15m
{
  "chart": {
    "result": [
      "meta": { "symbol": "KET NS" "exchangeTimezoneName": "Asia/Kolkata"
      "timestamp": 1788255000 1
      "indicators": { "quote": { "open": [ 1 "high": [ 1
      "low": [ 1 "close": [ 1 "volume": [ 111
    ]
  ]
}
```

The parser converts the parallel arrays into a table indexed by timezone aware timestamps and drops rows whose close is missing. For example, the cached 1 September 12:45 row becomes open 5,395.50, high 5,402.50, low 5,375.50, close 5,379.00, volume 25,558.

2.1 Storage refresh and data checks

Raw HTTP responses are cached under hashed filenames. Settled NSE files are cached indefinitely. Parsed daily rows are stored in data/asymmetry.db. A unique index on date and symbol blocks duplicates. Refreshing a date deletes and rewrites that day. The Yahoo parser drops bars without a close. The HTTP client paces calls, retries temporary failures, and stops retrying on 403 or 404.

The current validation layer is incomplete. It does not run a formal missing candle audit, enforce expected 15 minute timestamps, compare adjacent closes with previous close, adjust for splits or dividends, reconcile Yahoo with NSE, or record a data checksum and retrieval timestamp in every signal. The absence of those checks is a planned gap, not an implicit pass.

Local evidence on 10 September 2026	Value
SQLite daily rows	682,818
Daily date range	9 July 2025 through 9 September 2026
Distinct daily sessions	290
Distinct symbols in daily store	2,959
Catalyst rows	620
Legacy journal rows	11
V3 JSONL forward records	33, all still marked open in the inspected file
Run log directory	Not present

2.2 Frozen universe and sample

The file data/universe-2026-07-13.json contains 1,034 symbols. It was created from all EQ series symbols that passed the existing 20 session liquidity rule on 13 July: median turnover at least INR 5 crore, median volume at least 50,000, and trading on at least 80 percent of those sessions. It is not a historical NIFTY 500 membership file.

The sample function uses Python random.Random with seed 1 and draws 150 symbols uniformly without quality ranking. Reusing the same snapshot, sample size, and seed returns the same names. KEI is in the full frozen list and in the stated 150 name experiment. Freezing membership before the evaluation period reduces survivor and current universe bias, but it does not repair candle time leakage inside a replay.

3 Candles and timeframes

Timeframe	How it is obtained	Timestamp meaning	Safe availability rule
Daily	Official NSE stored rows or Yahoo direct chart	Trading date or 09:15 start depending source	Only after the session has closed and the daily record is complete
Weekly	Daily rows resampled with high max, low min, close last, volume sum	Week bucket	Only from completed daily rows
60 minute	Yahoo 60m or Upstox 1m resampled	Bar opening time	Available after start plus 60 minutes
120 minute	Pairs of consecutive 60m bars within each session	First 60m bar opening time	Available only after both component bars have completed
15 minute	Yahoo 15m or Upstox 1m resampled	Bar opening time	Available after start plus 15 minutes

NSE regular cash trading is represented as 09:15 through 15:30 IST. Yahoo returns 26 rows: 25 fifteen minute intervals and an extra row stamped 15:30 that the code treats as a closing print. The live scanner removes a currently forming candle. The historical backtest correctly removes the current daily candle, but its 60 minute slice uses bar start time rather than bar completion time. That is the source of the KEI carry leak.

4 Calculations used by V3

4.1 EMA ATR ADR and VWAP

Calculation	Exact implementation	Why it matters
EMA	pandas ewm with adjust False. New EMA = alpha x close + (1 - alpha) x prior EMA, alpha = 2 divided by span plus 1.	20 and 50 period alignment supports trend and carry. A 200 EMA is also used in the broader regime helper when enough history exists.
True range	Maximum of high minus low, absolute high minus prior close, and absolute low minus prior close.	Includes overnight gaps.
ATR 14	Wilder style exponential mean of true range with alpha 1 divided by 14.	Expressed as percent of price for move feasibility and other structure measures.
ADR 20	Mean of 100 x (daily high - daily low) divided by daily low over the last 20 rows.	Estimates the distance normally travelled inside a session.
VWAP	Session cumulative typical price times volume divided by cumulative volume; typical price is (high + low + close) divided by 3.	Implemented as a shared indicator, but the audited V3 selection, geometry, and score do not use it.

KEI numerical example. Using completed daily data before 1 September gives ADR 2.805 percent and ATR about 2.722 percent. Including the full 1 September daily candle raises ATR to about 3.299 percent. The supplied 3.31 percent therefore identifies the full day calculation, which was not available at 12:45.

4.2 Relative strength and sector leadership

For each horizon of 5, 20, and 60 sessions, excess return equals stock return minus benchmark return. The code averages those three excess returns, ranks every stock from 0 to 100, and repeats the process against its equal weighted sector composite. Acceleration is 5 session excess versus NIFTY minus 60 session excess. A separate composite score uses 40 percent stock versus NIFTY percentile, 35 percent stock versus sector percentile, and 25 percent acceleration percentile. Missing percentiles are filled with 50.

Sector leadership uses 20 and 60 session excess return of the sector composite versus the all stock benchmark. Acceleration equals 20 session excess minus 60 session excess. A sector is Leading when current excess and acceleration are positive, Improving when current excess is nonpositive but acceleration is positive, Weakening when current excess is positive but acceleration is not, and Lagging otherwise. The score is its cross sectional 20 session percentile. For shorts, stock and sector percentiles are mirrored as 100 minus the long percentile.

4.3 Setup rules

Setup	Rule	Quality score
Liquidity sweep	Find prior pivot shelf, require recent trade beyond it and latest close back through it. Implemented symmetrically for long and short.	45 percent shallow sweep, 25 percent number of touches, 30 percent reclaim distance.
High tight flag	At least 8 percent impulse over 20 bars, then 8 bar flag no deeper than half the impulse, close retains at least 60 percent of impulse.	40 percent tightness, 35 percent retention, 25 percent ATR contraction.
Base breakout	Eight preceding bars form a base no deeper than 8 percent. Final bar closes beyond the base on at least 2.0 times mean base volume.	35 percent tightness, 35 percent log scaled volume surge, 30 percent close location in direction of trade.

The detector evaluates all three and returns the found setup with the highest quality. It returns only one best setup for the stock. The daily scanner evaluates long and short and again keeps the higher quality direction. A later setup filter normally restricts production output to reclaim and base breakout because continuation measured poorly.

4.4 Structure headroom and carry

A pivot high or low is a bar that is the unique extreme across a centered window. Weekly and daily trend are higher high and higher low, lower high and lower low, or mixed. The stop uses the nearest 15 minute pivot on the invalidation side that is at least 0.5 percent from entry. It is never tightened to force qualification.

Carry combines five 60 and 120 minute components: 30 percent best 120 minute V3 setup quality, 20 percent 20 and 50 EMA alignment, 20 percent headroom, 20 percent volume sequence, and 10 percent range location. Volume sequence compares the mean of the last three 60 minute volumes with the preceding 12 bar base and adds credit when the second half of the base is quieter than the first. Headroom scores the distance to the next opposing 120 minute pivot relative to the required target move.

Only 120 minute structure present and volume sequence are named core checklist conditions in the current code. The total must also reach 60. Headroom is shown and scored, but it is not a veto. That is why the retrospective KEI carry record could pass at 62.2 despite failing headroom.

4.5 Quality score and missing data

Module	Weight	KEI reported module	Weighted points
RS versus NIFTY	12%	58.7	7.04
RS versus sector	12%	67.9	8.15
Sector leadership	10%	5.3	0.53
Weekly and daily structure	12%	22.0	2.64
Daily setup quality	10%	47.4	4.74
Entry quality	8%	60.4	4.83
Catalyst	12%	50.0	6.00
Volatility feasibility	6%	44.1	2.65
Carry	18%	62.2	11.20
Total	100%		47.78

The total is a weighted arithmetic average. It is not a probability. The 72, 76, and 80 regime floors were chosen from one observed live scan of 269 candidates whose scores ran from 55 to 81 with a median of 65. The repository calls this calibrated, but it is a manually selected throughput threshold rather than a statistically validated optimum.

Missing relative strength values are neutralized to 50 in the composite. Missing 60 minute history fails carry closed. Missing catalysts normally produce neutral 50 unless an optional catalyst hard gate is armed. An empty catalyst result across all candidates is treated as an outage and the gate is disarmed. These treatments are implemented choices, not learned rules.

5 How the scanner finds trades

1. Load up to 400 calendar days of stored daily rows ending on the requested date.
2. Load current NIFTY 500 membership, then apply the 20 session liquidity gate.
3. Build equal weighted market and sector composites and relative strength percentiles.
4. For each stock, detect the best daily long and short setup and keep the higher quality one.
5. Reject a direction opposed by weekly or daily trend. Run a cheap best case move feasibility check.
6. Order survivors by a cheap promise score. The configured maximum can truncate the paid intraday pass; zero means all.
7. Optionally collect catalysts. Run 60 and 120 minute carry before downloading 15 minute data.
8. Build the newest qualifying 15 minute plan in the last 26 bars. Apply stop, feasibility, structure, liquidity and technical hard filters.
9. Compute the nine module quality score and compare it with the regime floor.
10. Apply portfolio and daily caps, annotate probabilities when available, then write reports and the forward record.

The after close scheduled job normally runs once on weekdays. The live monitor can loop shortly after each 15 minute boundary, but it still starts from daily setups already in stored history. It does not build a developing daily base breakout from partial intraday data. Repeated signals are limited by one best daily setup per stock, a recent plan search, daily caps, portfolio caps, and replay busy until logic that avoids overlapping trades in one symbol.

6 From setup to trade plan

For a reclaim, confirmation is considered complete and the quoted entry is the current close, with replay entry on the next bar open. For continuation and base breakout, the order is a stop through the lowest prior session level for a short or the highest prior level for a long, with a 0.05 percent buffer. For KEI short, the prior 15 minute low is 5,387.50 and the buffered price is $5,387.50 \times 0.9995 = 5,384.81$.

KEI value	Calculation
Entry 5,384.81	$5,387.50 \times 0.9995$, rounded to two decimals
Stop 5,460.00	Nearest qualifying 15 minute pivot high at least 0.5 percent above entry
Risk 75.19	$5,460.00 - 5,384.81$
Stop distance 1.396%	75.19 divided by $5,384.81 \times 100$
Valid fill minimum 5,379.31	$5,460$ divided by 1.015
Valid fill maximum 5,432.84	$5,460$ divided by 1.005
3R target 5,159.24	$5,384.81 - 3 \times 75.19$
4R target 5,084.03	$5,384.81 - 4 \times 75.19$

KEI value	Calculation
Required move 5.59%	Absolute target divided by entry minus 1

The code rounds levels to two decimals, not to an exchange tick size. No tick table is implemented. An unfilled stop order waits up to four later 15 minute bars and is cancelled at the session close. A fill outside the valid band is rejected because the fixed structural stop would no longer be between 0.5 and 1.5 percent away.

Move feasibility uses capacity = minimum positive value of ADR and ATR, multiplied by the square root of five sessions. It accepts when required move divided by capacity is at most one. With pre 1 September inputs, capacity is $\min(2.805, 2.722) \times \text{square root of } 5$ = about 6.09 percent, so 5.59 percent passes. The reported 3.31 percent ATR and 44.1 volatility score use the completed 1 September daily bar. That is an unavailable input at 12:45 even though the accept or reject outcome happens to remain the same.

Quantity is integer risk budget divided by risk per share. With no account equity configured, the legacy risk budget is INR 5,000, so KEI sizes to 66 shares and about INR 355,397 notional. The code can cap position value, concurrent positions, portfolio heat, and positions per sector when account equity is configured.

Instrument status is not verified. The plan uses NSE cash share prices and integer shares. It does not choose a stock future, fetch futures lot size, reserve margin, prove stock borrow, model funding, or calculate instrument specific fees. An overnight short therefore cannot be called executable from these files.

7 Filtering scoring and publication

Layer	Question	Current effect
Setup	Does a permitted daily pattern exist	No pattern means no candidate
Geometry	Is the structural stop 0.5 to 1.5 percent and is the target feasible	Hard rejection
Carry	Does a base breakout or continuation have sufficient 60 and 120 minute support	Hard technical rejection for gated setups
Quality	How strong are the nine modules	Weighted score only
Publication	Does score clear 72, 76, or 80 and fit portfolio and daily caps	Trade, watch, or demotion
Probability rollout	Does untouched evidence and live data permit an alert	Current model keeps funded alerts disabled

KEI was in a selective regime, so the reported score of 47.78 was below 76. It was not published on the 1 September V3 brief, which showed zero setups. Failed headroom did not veto the trade because headroom is not in the carry core condition list and weekly structure is the only opposing level that can block geometry. Sector leadership also scores rather than gates.

The rules came from a mixture of engineering judgment, prompt specifications, later measurements, and bug fixes. The carry setup and production setup restrictions have cohort measurements in repository notes. The regime score floors and many formula weights are manually selected. They must be treated as unvalidated design choices until tested on untouched and forward data.

8 Where AI is involved

Stage	Method	AI input and output	Control
Prices, indicators, setup, stop, target, replay	Fixed Python rules and arithmetic	No AI	Unit tests and deterministic inputs
Catalyst interpretation	LLM cascade across configured Cerebras, Groq and Gemini compatible endpoints	Headline plus context in; strict JSON catalyst fields out	Clamp ranges, parse JSON, fall through on failure, neutral fallback
Procedural filings	Fixed routing rules	No AI for skipped or neutral event records	Subcategory route
Results PDFs	PDF extraction plus optional LLM	Extracted figures and filing headline in; catalyst JSON out	Neutral if PDF or model cannot be read
V3 report writing	Deterministic Python strings and HTML templates	No AI	Numbers copied from structured scan object
This audit document	Human directed document generation based on repository evidence	Repository files and attached brief in; explanatory document out	Claims separated by evidence status

The catalyst system prompt asks whether news changes future earnings or intrinsic value, not whether it sounds positive. The output includes catalyst type, expectation delta from -3 to 3, materiality, durability, already priced, confidence, and a short rationale. Model temperature is 0.1. Repeated model calls can still differ because external LLM services are not guaranteed deterministic. Cached catalyst rows make later runs stable once stored, but provider and prompt changes can change new results.

9 How replay measures outcomes

The replay first forms a signal from data sliced to the decision point, then calls a next bar entry simulator. Once filled, it walks each 15 minute bar forward for five sessions. For a long, low at or below stop is a stop and high at or above target is a target. For a short, high at or above stop is a stop and low at or below target is a target. The stop check happens first, so a candle touching both levels is booked as a loss.

It separately records whether 2R, 3R, and 4R were touched. A requested fixed target ends the simulated trade at that target. Therefore 3R and 4R are separate fixed exit scenarios or labels, not a coded partial exit plan. If neither level is reached, the position is marked to the final horizon close. Maximum favorable excursion and maximum adverse excursion are stored in R units.

Transaction cost per trade is modeled as (0.12 percent round trip plus 0.05 percent slippage) divided by the trade stop percent. This is an estimate. It is not a broker contract note and does not include a verified short instrument, margin, borrow, gap slippage distribution, or futures basis.

On 4 September the KEI 09:15 bar opened at 4,995, below the 4R target of 5,084.03. The worked replay books exactly 4R at the target. That is a conservative capped payoff for comparability, but it is not a realistic fill model. A market or stop exit already resting through a favorable gap might fill nearer the open; a limit target has different mechanics. The code does not identify an order type for the exit, so exact executable proceeds are not verified.

10 How final documents are generated

The scan builds a V3Scan object containing regime, source tier, funnel counts, threshold, trade candidates, near misses, rejection counts, probability status, and rollout status. Each candidate holds the setup, plan, module scores, carry checklist, catalysts, and explanatory notes. The Markdown builder and HTML builder read those records and place them into fixed text templates. The terminal renderer uses the same object.

No LLM writes the V3 brief. This reduces invented prose, but traceability is still incomplete because output statements do not carry source record identifiers, data hashes, calculation versions, or an immutable decision timestamp. A reader can trace many fields back to code and cached candles manually, but not by following a first class provenance link.

KEI appeared in a retrospective worked example because a later search examined the 1 through 4 September price path and identified a successful geometry. The generated 1 September V3 brief had published zero trades. The repository does not contain a preserved run manifest listing every short candidate considered by that winner search, its exact command, code commit, cache hashes, or rejected cohort. The example is therefore a useful forensic case and an invalid selection performance claim.

11 Reproducing the current process

11.1 Requirements

- Windows, macOS, or Linux with Python 3.11 or later and the project dependencies installed.
- Network access to NSE archives and Yahoo for a fresh run. BSE and an LLM key are optional for catalyst scoring.
- An Upstox API application and daily access token for authenticated live monitoring. The repository holds market data methods only.
- A writable data directory for SQLite, HTTP cache, reports, JSON models, and JSONL forward records.
- Account equity is needed before the rollout can call any position live ready. It is not configured in the inspected environment.

11.2 Commands

```
# install the project
uv sync
# download and store recent official daily data
uv run acvmmetry backfill --days 1000
# generate the normal brief and web page
uv run acvmmetry brief --html
# run the current V3 after close
uv run acvmmetry v3 --no-refresh --per-day 3 --setup reclaim --setup base-breakout
# reproduce the frozen universe replay from cached bars
uv run asymmetry v3-backtest --symbols-file data/universe-2026-07-13.json --sample 150 --seed 1 --score-modules --cache-only
# fit and check the probability model
uv run acvmmetry v3-calibrate --symbols-file data/universe-2026-07-13.json --sample 150 --seed 1 --cache-only
# run one live paper monitor pass during market hours after refreshing Upstox auth
uv run acvmmetry v3-monitor --once
# run tests
python -m pytest -q
```

The inspected test run completed with 294 passed and 1 skipped. This proves the encoded tests pass; it does not prove the financial strategy works or that the KEI short replay is correct, because the important short and bar completion cases are not fully represented by those tests.

The current historical replay is a batch study over cached bars. A functioning live scanner also needs exchange aligned closed bar handling, authenticated live data, an intraday candidate state that does not rely on the future daily close, immutable alerts, alert delivery, operational logs, health monitoring, and instrument aware execution assumptions. These are roadmap items.

12 Complete KEI audit

12.1 What the cached prices verify

Time IST	Open	High	Low	Close	What can be said
1 Sep 12:15	5,420.00	5,422.50	5,387.50	5,393.00	Creates the 5,387.50 prior low
1 Sep 12:30	5,395.00	5,410.00	5,393.00	5,398.00	No entry crossing
1 Sep 12:45	5,395.50	5,402.50	5,375.50	5,379.00	Bar crosses 5,384.81, but this is the signal bar
1 Sep 13:00	5,378.00	5,395.00	5,371.50	5,389.00	First bar after signal; opens below sell stop
1 Sep after 13:00		5,420.00	5,310.00	5,348.00	Stop at 5,460 untouched
2 Sep 14:00	5,165.00	5,183.50	5,154.00	5,158.00	3R target 5,159.24 traded
4 Sep 09:15	4,995.00	5,025.00	4,863.00	4,975.00	Gap below 4R target 5,084.03

The price path supports a hypothetical short entered near the quoted level with the quoted stop. It does not by itself prove that the system knew to place the order before the move.

12.2 When the signal was knowable

1. Yahoo stamps the 12:45 candle at its opening time. Its close, low, high, and volume are known only at about 13:00.
2. A rule calculated from the completed 12:45 bar can place an order no earlier than the next bar. The 12:45 low cannot be used as proof of fill for an order created after that bar closed.
3. The first tradable bar opens at 5,378.00, below the sell stop at 5,384.81. A realistic short stop fill would need an explicit gap rule and would usually not receive the better 5,384.81 price automatically.
4. The current simulator checks open greater than or equal to entry, then high greater than or equal to entry for every stop through order. Those are buy stop comparisons. It returns 5,384.81 for KEI because the next bar high was 5,395, even though the direction is short.

12.3 Daily lookahead

Daily information supplied to detector	Detector result
Completed data through 31 August, which is all that existed at 12:45 on 1 September	No short setup. The detector reports a different prior level at 5,233 and says it was not rejected.
Full 1 September daily high, low, close, and volume	Short base breakdown: 8 bar base, 5.8 percent deep, level 5,442, 2.3 times volume, close location 85 percent.

The sentence about an 8 bar base is therefore a description made with the final daily candle. At 12:45 the session still had 2 hours and 45 minutes left. Its final close and full day volume were future information. This invalidates the claim that the daily scanner selected KEI at 12:45.

12.4 Higher timeframe lookahead

60 minute cutoff	Carry score	Result
Bars whose start timestamp is at or before 12:45	62.2	Passes, but includes the 12:15 to 13:15 candle before it completed
Only bars completed by 12:45	45.2	Fails the 60 point floor

The 62.2 score is composed of 120 minute setup 44.2, EMA alignment 100, headroom 8.8, volume sequence 87.5, and range location 96.5. The completed only score has 120 minute setup 0, alignment 100, headroom 14.9, volume sequence 63.1, and range location 96.2. The historical slice condition hourly index less than or equal to signal time treats a start timestamp as if it were a completion timestamp.

12.5 Score and publication

The arithmetic that produces 47.78 is correct for the supplied module values. Several inputs are not point in time: the setup quality 47.4 and volatility score 44.1 depend on the completed 1 September daily candle, while carry 62.2 depends on the incomplete 60 minute candle. Correct arithmetic does not make unavailable inputs valid.

Even using those retrospective inputs, 47.78 is below the selective floor of 76, so the V3 publication decision is rejection. The 1 September generated V3 report published zero trades. The worked KEI result was selected later because its subsequent path was attractive. The repository does not preserve the full winner search cohort, so winner selection bias cannot be measured from this example.

12.6 Executability and outcome

A hypothetical fill at 5,384.81, stop at 5,460, and fixed targets produces 3R at 5,159.24 and 4R at 5,084.03. The cached candles touch both targets before the stop. Capping the favorable gap at exactly 4R produces a comparable research label. It is not evidence of a broker fill, because the repository has no order record, broker statement, chosen instrument, lot size, margin, borrow, funding, or instrument specific transaction costs.

Audit verdict. KEI is a valid worked example of geometry and subsequent market movement. It is not a valid prospective V3 discovery example. It should remain in documentation only as a counterexample that demonstrates why event time and direction aware fill tests are necessary.

Part Two Evidence Appendix

A Evidence inventory

Evidence	Status	What it establishes
Attached seven page engineer brief	Reviewed	Requested teaching document, KEI audit, evidence appendix, and roadmap
src/asymmetry	Inspected	Actual data, indicator, setup, carry, V3, replay, AI, report, and CLI logic
data/asymmetry.db	Queried read only	Daily, catalyst, and journal row counts and date coverage
KEI Yahoo cache files	Replayed read only	Daily, 15 minute, 30 minute, 60 minute and 5 minute cached data exist
data/briefs/2026-09-01-v3.md	Reviewed	Zero published setups and selective floor 76
docs/2026-09-09-last-week-positive-trade.md	Audited	Source of KEI claimed figures; point in time claim does not survive reproduction
docs/2026-09-09-oos-universe-measurement.md	Reviewed	Frozen universe cohort results and limitations
docs/2026-09-10-v3-corrected-probability-report.md	Reviewed	Current decision is paper only and no approved funded strategy
pytest	Executed	294 passed and 1 skipped in 13.29 seconds

B Code reference map

Topic	Reference
Settings, weights, thresholds, costs and capital	src/asymmetry/config.py
NSE archive formats	src/asymmetry/data/nse_archive.py
Yahoo timestamps and parsing	src/asymmetry/data/yahoo.py
Upstox read only live data	src/asymmetry/data/upstox.py
Cache pacing and retry	src/asymmetry/data/cache.py
SQLite storage and unique indexes	src/asymmetry/storage.py
Universe freeze and seeded sample	src/asymmetry/universe_snapshot.py
EMA ATR ADR VWAP	src/asymmetry/engines/indicators.py and v3.py lines 58 to 92
Setup rules	src/asymmetry/engines/setups.py
120 minute construction and carry	src/asymmetry/engines/carry.py lines 143 to 340
Trade geometry and quality score	src/asymmetry/engines/v3.py lines 292 to 598
Stage ordering and score floor	src/asymmetry/engines/v3_scan.py lines 191 to 603
Replay and entry fill	src/asymmetry/v3_backtest.py lines 356 to 674
Catalyst prompt and JSON	src/asymmetry/intelligence/prompts.py and openai_compat.py
Markdown and HTML reports	src/asymmetry/report/v3_report.py and v3_website.py
Scheduling	scripts/daily_run.ps1 and .github/workflows/daily-brief.yml

C Verified defects and gaps

Priority	Finding	Consequence	Required proof of fix
Critical	Daily 1 September candle used to label KEI before that candle closed	Future data changes no setup into a base breakdown	Decision snapshot excludes the current daily candle during the session
Critical	Historical 60 minute slice uses bar start time as availability time	Carry rises from 45.2 fail to 62.2 pass	Every bar carries start and available at timestamps; tests cover exchange anchored completion
Critical	Stop through fill logic is not direction aware	Short gap fill can receive an impossible better price	Short sell stop tests for open below trigger and low crossing; gap fill policy documented
Critical	Main backtest is long only	KEI short cannot be reproduced by the advertised command	Direction parameter and symmetric fixtures for long and short
High	No instrument model for overnight shorts	Quantity and costs may not be executable	Instrument, lot, margin, borrow or futures basis and costs stored in plan
High	No immutable run manifest or full candidate ledger	Retrospective winner search cannot be audited	Every decision and rejection written before outcomes with code and data hashes
High	No missing candle or corporate action audit	Bad inputs can create false setups and stops	Data quality report blocks scan on unresolved anomalies
Medium	Tick size rounding is absent	Displayed prices may not be valid orders	Instrument tick rule applied to every level
Medium	No run logs present in inspected workspace	Operational failures and exact commands are not reconstructable	Structured logs retained with run IDs and status

D Current statistical evidence

The 9 September frozen sample study reported 5,478 triggers before the later probability correction. The carry admitted blend was positive, but continuation remained negative, base breakout had only 22 gated examples, and reclaim carried most of the sample. The 10 September corrected probability report supersedes any funding implication: no strategy cleared all gates, the final usable window was only three sessions, and funded alerts remain disabled.

Corrected final result	Raw final	Conservative	Decision
Reclaim 2R	7 of 24 or 29.2%	0.0% and -1.24R conservative net	Paper only
Reclaim 3R	6 of 24 or 25.0%	0.0% and -1.24R conservative net	Paper only
Reclaim 4R	6 of 24 or 25.0%	0.0% and -1.24R conservative net	Paper only
Base breakout 2R 3R 4R	0 final observations	Unavailable	Paper only

E Claims that remain unverified

- A prospectively knowable KEI short signal at 12:45 on 1 September.
- A broker executed KEI entry, exit, cost, margin, borrow, or futures position.
- The complete search population from which the later KEI winner was selected.
- Current external provider pricing, entitlements, and service level commitments.
- Corporate action corrected intraday history and full historical catalyst coverage.
- A stable out of sample probability of reaching 3R or 4R after all costs.
- A funded rollout approval. The current files say paper only.

Part Three Practical Build Roadmap

13 Inventory and first version

Category	Already exists	Broken or unverified	Missing
Data	NSE daily store, Yahoo cache, optional Upstox, timezone aware frames	Completion time, missing candle and corporate action validation	Canonical bar contract with provenance and quality status
Signals	Three daily setups, carry, trade geometry, quality score	Short replay and developing daily pattern mismatch	Prospective state machine and direction symmetric tests
Replay	Next bar flow, stop first ambiguity rule, fixed targets, R metrics	Short fill and higher timeframe leakage	Order type specific gap and slippage model
Selection	Regime floor, nine modules, probability gate	Several manual weights and tiny final sample	Pre registered evaluation and full rejection cohorts
Operations	CLI, GitHub Actions, HTML and Markdown output	No inspected run logs and open old JSONL records	Immutable alerts, delivery, health checks and reconciliation
Execution	Risk budget arithmetic only	Instrument and costs not executable	Separate future stage for broker and instrument integration

13.1 Recommended first version

- Universe: the frozen liquidity eligible EQ universe, with a new point in time snapshot per study. Do not use current NIFTY membership to select old tests.
- Direction: long only for the first repair release, because the production report and calibrated replay are currently long only. Add shorts only after instrument and fill semantics pass dedicated tests.
- Instrument: cash equity paper trades only. No overnight short execution in version one.
- Setups: reclaim and volume backed base breakout. Keep high tight flag disabled until a controlled redesign shows positive net evidence.
- Timeframes: previous completed daily and weekly context, completed 60 and 120 minute carry, completed 15 minute trigger.
- Holding period: one to five completed exchange sessions with fixed 2R, 3R, and 4R evaluation labels. Do not mix those labels into one exit plan.
- Alert timing: five seconds after a 15 minute close, only when every input has available_at less than or equal to decision_at.
- Output: timestamped paper alert plus full eligible, rejected, unfilled, expired, and settled event records.

Owner decisions still needed are account size for later sizing, acceptable data vendors and budget, whether the first live study is long only, which instrument will represent shorts in a later stage, the alert channel, acceptable trading frequency, and risk limits for a future funded pilot. These choices should not be inferred from KEI.

14 Implementation sequence

Stage	Deliverable and dependencies	Effort estimate	Recurring cost estimate	Objective acceptance check
0 Integrity repair	Bar availability contract; direction aware fills; depends on cached fixtures	4 to 6 engineer days	None	KEI no longer appears prospectively at 12:45; carry is 45.2 with completed bars; short gap test fills by documented policy
1 Data validation	Session calendar, expected bar grid, duplicate, timestamp, price, volume and corporate action checks	5 to 8 days	Public sources may be zero; licensed corporate action feed to quote	At least 98 percent expected bars present; unresolved anomalies block a signal; daily reconciliation report saved
2 Deterministic calculations	Versioned indicator and setup library with golden fixtures	4 to 6 days	None	Independent hand calculations and code agree within tick or 0.01 percent tolerance
3 Prospective detector	Closed bar state machine for reclaim and base breakout	5 to 8 days	Live market data plan to confirm	Replaying event stream and running live stream produce identical decisions
4 Geometry and replay	Instrument neutral order model, next bar fills, gaps, stops, targets and costs	5 to 7 days	Broker and exchange fees must be refreshed	Long and short fixtures pass; no same candle trigger and fill without finer data
5 Candidate ledger	Append only event store for every candidate and rejection with hashes	4 to 6 days	Local SQLite near zero or managed storage quote	A report statement resolves to raw bars, version, formula, and decision event
6 Selection validation	Geometry only versus full publication paired evaluation	5 to 8 days plus data collection	Compute only	Untouched groups meet pre registered sample, probability and net R gates
7 Paper alerts	Alert schema, delivery, acknowledgements and original payload preservation	4 to 6 days	Messaging service likely zero to modest; quote before use	Thirty consecutive forward signals reconcile without rewrite, loss, or duplicate
8 Monitoring and reports	Health dashboard, stale feed alarm, settlement, cost sensitivity and cohort reports	4 to 6 days	Hosting estimate INR 0 to 5,000 monthly until vendor quotes	Daily run manifest complete; failures visible; all open events settle or explain why
9 Live execution later	Separate broker, instrument, margin and kill switch specification	Not estimated until owner decisions	Brokerage, data and margin are vendor specific	Requires separate approval; not part of immediate milestone

Effort and recurring costs are planning estimates, not verified quotes. Current vendor plans and exchange requirements can change. Obtain written prices and entitlements before committing to a service.

15 Immutable paper alert example

The first useful alert record is machine readable and never edited. Later outcomes are appended as separate events. The following KEI record is intentionally rejected; it shows how the repaired system should preserve the evidence instead of rewriting history.

```
event_type: candidate rejected
event_id: 2A26A00A1T13A0A05+A053A KEI short
instrument: KEI MFE cash reference
decision_time: 2A26-A0-A1 12:00:05 TST
data_cutoff: 2A26-A0-A1 12:00:00 TST
direction: short
setup_requested: base breakdown
entry_rule: sell stop through 5387 50 with a 05 percent buffer
planned_entry: 5384 81
structural_stop: 5460 00
targets: 3R 5150 21 1R 5084 03
valid_fill_band: 5370 31 to 5422 84
entry: four completed 15 minute bars or session close
status: REJECTED
reason 1: no completed daily short setup at decision time
reason 2: completed bar carry score 45.2 below 60
code_version: required commit hash
data_hashes: required raw response hashes
payload_hash: required SHA256 over the original alert body
```

A qualifying alert uses the same schema with status PAPER ALERT and includes the exact passed rule values. Its settlement event later records fill, stop, target, timeout, costs, MFE, MAE, and evidence links without changing the original payload.

16 Validation design

The next study must log every eligible geometry, including losses, unfilled orders, expired orders, score rejections, missing data exclusions, and portfolio demotions. Two policies are evaluated on the same event set: geometry only and the full publication model. This isolates whether scoring adds value instead of comparing different opportunity populations.

1. Freeze code, formula versions, universe rule, costs, setup definitions, timeframes, and evaluation criteria before producing evaluation outcomes.
2. Use a chronological development period for implementation and one round of rule selection.
3. Leave a five session embargo, then use a validation period to choose among already declared variants.

4. Leave another five session embargo and run one untouched final period exactly once.
5. Start forward paper monitoring with the chosen frozen version. Any rule change creates a new version and a new forward cohort.
6. Keep low score rejected base breakdowns as a complete cohort. Never use KEI alone to justify a threshold change.

Metric	Required reporting
Coverage	Eligible stocks, expected and received bars, missing data exclusions, corporate actions and source tiers
Opportunity counts	Signals, fills, unfilled, expired, score rejected, hard rejected and portfolio demoted
Outcome	Win rate, 2R 3R and 4R before stop or expiry, average win, average loss, gross and net R per trade
Risk	Maximum drawdown in R, losing streak, MAE, MFE, gap loss and portfolio heat
Uncertainty	Exact sample size, number of symbols and sessions, Wilson interval and bootstrap net R interval
Conditions	Results by setup, direction, regime, stop band, time of day, sector and data tier
Costs	Base, low and high cost assumptions; slippage by liquidity and gap condition
Model comparison	Paired geometry only versus full publication on identical opportunities

16.1 Criteria fixed before the next final test

- Zero signals may use a bar whose available_at is after decision_at.
- Every exact setup, target, and direction group needs at least 100 untouched evaluation trades, 30 symbols, and 20 sessions before a funded probability claim. These match the current configuration minimums.
- The lower Wilson probability bound must exceed cost adjusted break even for the target, and conservative net expectancy must be at least plus 0.40R per trade, matching the current configuration.
- Forward paper monitoring needs at least 30 immutable signals before rollout review. More data may be required when intervals remain wide.
- The full publication model must improve net R or reduce drawdown relative to geometry only without depending on one symbol, one week, or one market regime.
- No funded rollout occurs while data quality, instrument feasibility, or reconciliation acceptance checks fail, even if historical returns look attractive.

Win rate alone is not an approval rule. A 4R plan has a 20 percent gross break even rate before costs, but the required rate rises with cost and gap loss. Report the full payoff distribution and uncertainty.

17 First development milestone

Milestone name Point In Time Replay Contract

The milestone delivers a versioned bar object with start, end, and available at times; direction aware order fills; a candidate and rejection ledger; KEI golden fixtures; and a reproducible command that emits a signed run manifest.

17.1 Files and access needed from the owner

- Permission to use the existing repository and cached KEI market data as regression fixtures.
- A decision to keep milestone one long only or to include a specific short instrument after its rules are documented.
- Authenticated Upstox market data access for a later market hours acceptance run. No trading permission is required.
- A chosen source for exchange holidays, corporate actions, and instrument tick and lot rules.
- The preferred paper alert destination and retention requirement.

17.2 Evidence that proves completion

1. The full test suite passes and new tests fail on the old time leak and short fill code.
2. At 12:45 or 13:00 on 1 September, the KEI fixture produces no prospective daily base breakdown and carry remains below 60 using completed hourly data.
3. If a synthetic short sell stop is active before a gap below the trigger, the fill follows the documented gap policy and never receives an impossible better trigger price.
4. A replay and an event by event stream over the same fixture produce byte equivalent decisions.
5. Every alert, rejection, fill, expiry, and outcome contains decision time, data cutoff, code version, formula version, and source hashes.

6. The run report reconciles all candidate states and leaves no silent exclusions.

Once that milestone is complete, the team can run a clean prospective experiment. Until then, additional threshold tuning would measure the current time and fill defects rather than the intended trading idea.

Final operating position

Use the current system for research, deterministic reports, and controlled paper experiments. Do not present KEI as a prospectively discovered trade, do not lower the quality threshold to include it, and do not enable funded alerts from the current probability report. Repair event time and fill semantics first, preserve every opportunity before its outcome, and judge 3R and 4R performance only on complete untouched cohorts.